Factor Modelling Theory

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Outline

1. Theory of a quant modelling
    a. CAPM
    b. Fama French 3 factor Model
    c. Momentum
2. Modelling Techniques
    a. Horizons, decays, normalization
    b. Box-Cox transform
3. Model Selection (mixing factors)
    a. Forward, backwards selection
    b. Penalization: LARS, Lasso, etc
    c. Gram–Schmidt process
    d. Neural Network

Note: In this Lecture I will use materials from lectures of John Cochran and Markus Brunnermeier

Why do we need Factors?

- Imagine an investable universe of 5000 companies
- How would you make an investment decision?
- Dimensionality reduction
- Scalability
- Single Factor Model (CAPM)
- Three Factor Model (Fama French)
- Multi Factor Models

Single Factor Model

$$(r_i - r_f) = \alpha_i + \beta_i(r_M - r_f) + e_i$$

a_i = alpha

$b_i(r_M - r_i)$ = the component of return due to market movements (systematic risk)

e_i = the component of return due to unexpected firm-specific events (non-systematic risk)

Single Factor Model

What is the meaning of this model?:

$$\boxed{(r_i - r_f) = \alpha_i + \beta_i(r_M - r_f) + e_i}$$

Think of it as:

$$E(R_i) = \beta_i * \lambda + \epsilon_i$$

Expected excess return of stock is defined by its beta and the expected excess market return

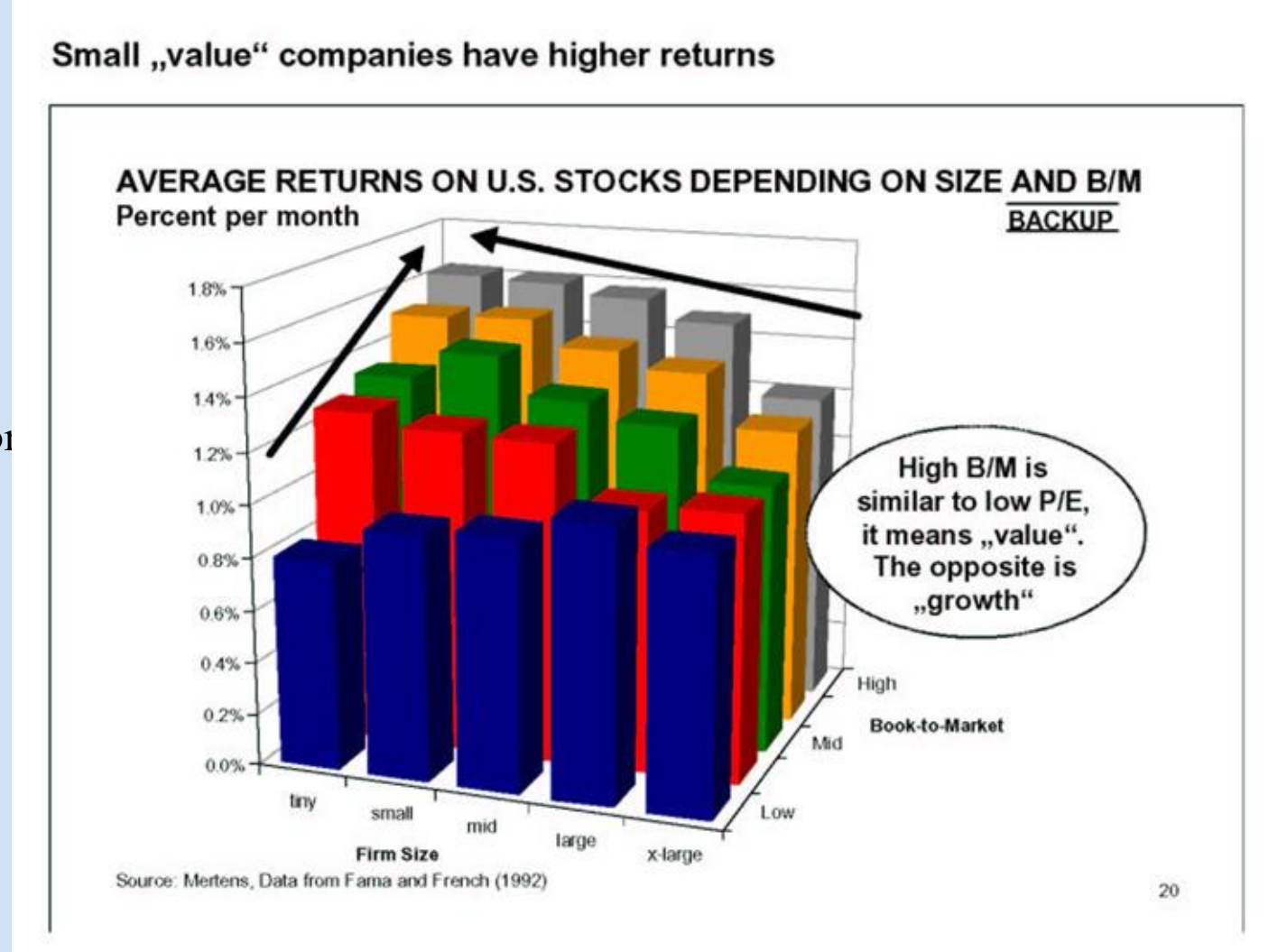
=> YOU CAN NOT DIVERSIFY

Fama French Model

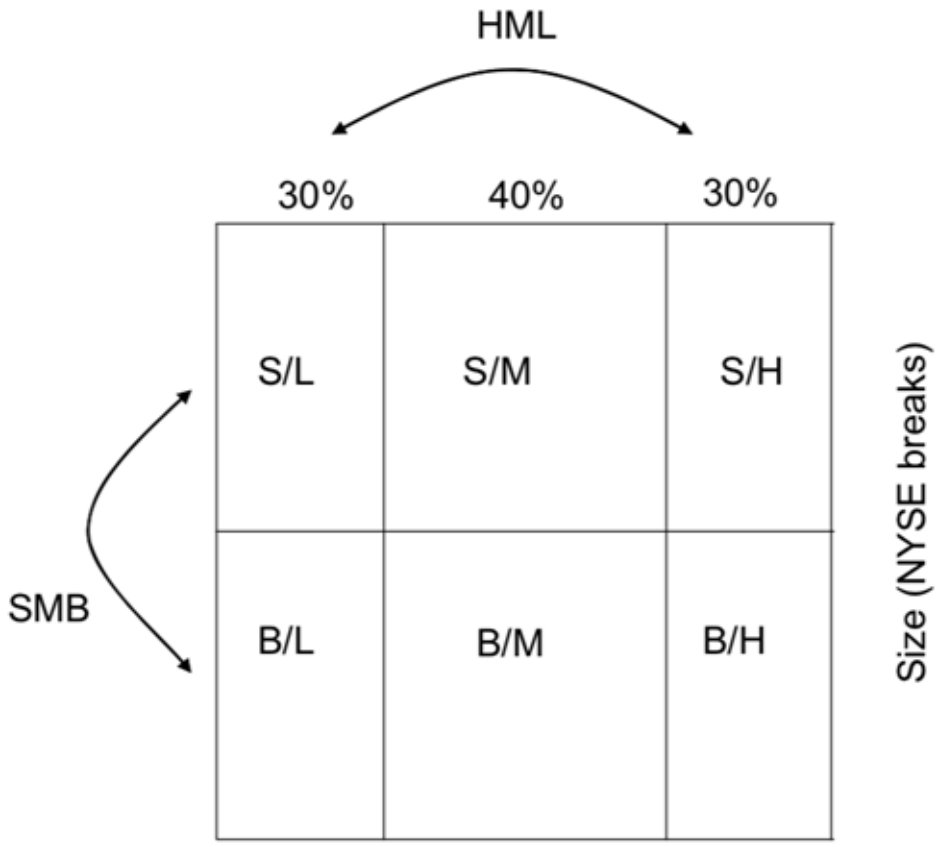
Observations:

- Small Names have higher expected returns
- High Book to Market (Value stocks) outperfor

  Low Book to Market (Growth Stocks)



Fama French Model



HML = (S/H + B/H)/2 – (S/L+B/L)/2

SMB = (S/L +S/M+S/H)/3 – (B/L+B/M+B/H)/3

Fama French Model

$$R_{i,t} - r_{f,t} = \alpha_i + b_{i1}(R_{m,t} - r_{f,t}) + b_{i2}SMB_t + b_{i3}HML_t + \varepsilon_{i,t}$$

SMB = “small [market capitalization] minus big”

"Size" This is the return of small stocks minus that of large stocks. When small stocks do well relative to large stocks this will be positive, and when they do worse than large stocks, this will be negative.

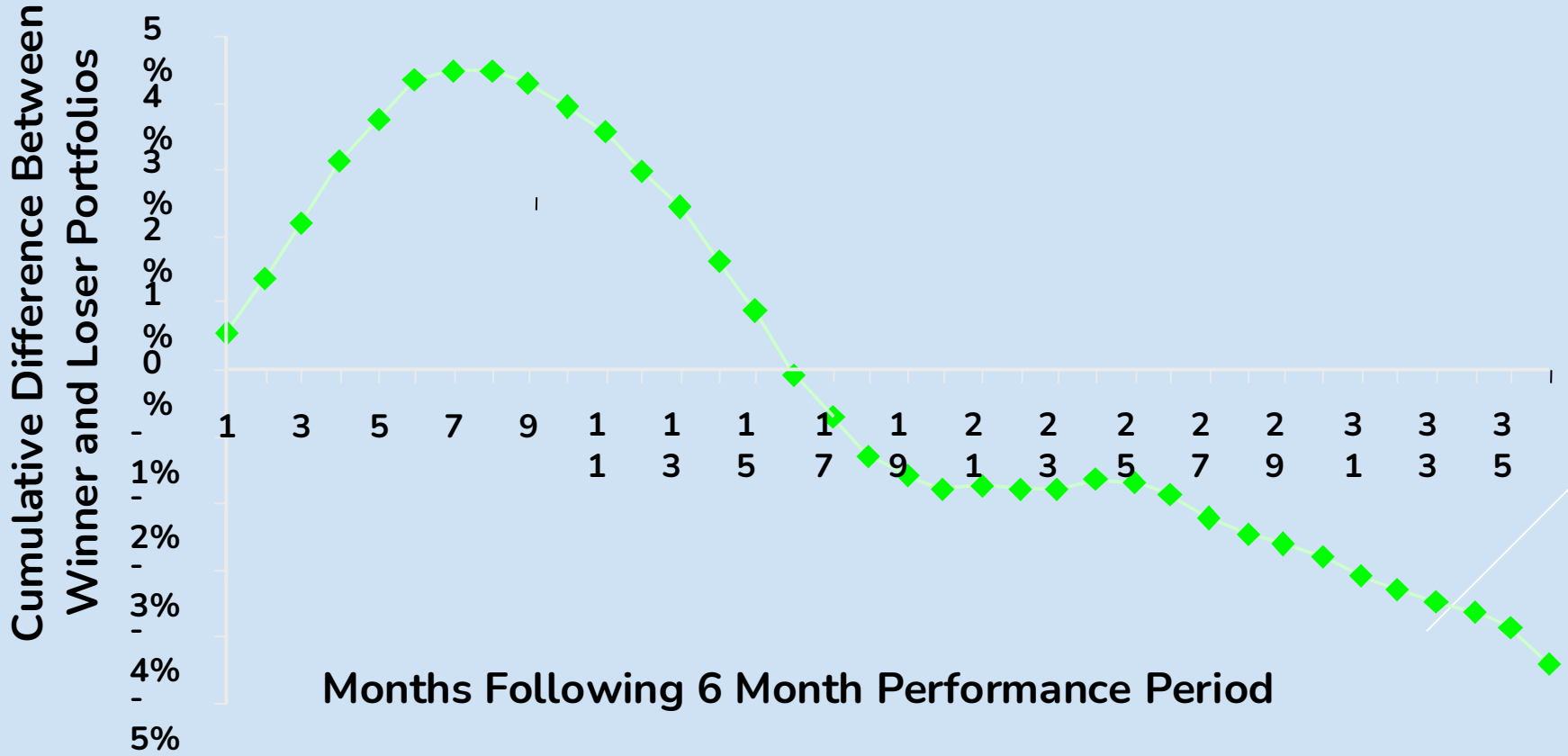
HML = “high [book/price] minus low”

"Value" This is the return of value stocks minus growth stocks, which can likewise be positive or negative.

Adding Momentum Factor

Momentum Factor: Top Winner minus Bottom Losers Portfolios over the last 6 months

Jegadeesh & Titman 1993 JF



Standard Quant Factors

- Book to Market
- Earnings to Price
- Dividend Yield
- Sales to Price
- Return on Equity
- Return on Assets
- Return on Investable Assets
- Accruals
- Sales to Total Asset Ratio
- Equity Ratio
- Total Asset Growth
- Investments Growth
- Investments to Assets Ratio
- EPS revision (1 month)
- EPS revision (3 month)
- Market Beta
- Momentum 1 month
- Momentum 12 months
- Volatility
- Skewness
- Trading turnover signed
- Trading turnover unsigned

Modelling a Factor - Statistics

Let's look at an simple OLS-regression:

$$E(R_i - R_f) = \alpha + \beta_i * F + \epsilon_i$$

Main requirements for a good fit:

- well distributed epsilons, theory says - second moment should exist and there should be stationary
- practice- the closer the shape of the distribution to the normal the better it is
- garbage in, garbage out

What can happen if F is not properly normalised:

- biased estimates, slow convergence, no statistical significance

Rule 1 of practical statistics: **always normalize factors before running regressions**...

How to Normalize Factors

1. sabtruct expected values and take logs
2. apply cox-box
3. check for normality of the distribution
    a. D'Agostino-Pearson
    b. Jarque-Bera
4. Let's look into the python code….

Model Selection

1. Forward Selection
2. Backwards Elimination
3. Regressions with penalties on the number of variables/ sum of absolute weights:
    a. Lasso (L1 norm) ALWAYS LASSO REGRESSION
    b. Ridge (L2 norm)
    c. LARS (General case)
4. Gram-Schmidt process
5. Neural Networks/ Support vector machines

Goal: avoid multicollinearity and include all possible available information

Model Selection(Forward Selection)

1. Start with a model with no predictors.
2. Add variable with largest F- statistic (provided P less than
3. some cut-off).
4. Refit with this variable. Recompute all F- statistics for adding one of the remaining variables and add variable with largest F- statistic.
5. Continue until no variable is significant at cut-off level.

Model Selection(Backwards Elimination)

1. Start with model with all predictors.
2. Delete variable with smallest F -statistic (provided P more than some cut-off).
3. Refit with this variable deleted. Recompute all F statistics for deleting one of the remaining variables and delete variable with smallest F statistic.
4. Continue until every remaining variable is significant at cut-off level.

Model Selection(Gramm-Schmidt)

$$F_1, F_2, \ldots F_N \text{ are the factors}$$

$$R_i \text{ is the return of stock } i$$

$$step1: \text{ find factor } k \text{ wuth the higested } F-\text{statistic}:$$

$$R_i = \alpha_i + \beta_i * F_k + \epsilon_i$$

$$step2: \text{ take residulas of the first regression for the factor } k:$$

$$\epsilon_i^{step1} = R_i - \alpha_i - \beta_i * F_k^{step1}$$

$$\text{and find a factor } k_2 \text{ such that regression}$$

$$\epsilon_i^{step1} = \alpha_i + \beta_i * F_{k2} + \epsilon_i^{step2}$$

$$\text{has the highest } F-\text{statistic}$$

$$step3 \ldots stepN$$

$$\text{continue as long as new factors are statistically significant}$$

Model Selection(LASSO)

$$min(\beta)\sum_{i=1}^{N}(R_i - \beta_0 - \beta_1 * F_1 - \beta_2 * F_2 - \ldots\ldots \beta_K * F_K)^2$$

$$st. \sum_{k=1}^{K}|\beta_k| < 1$$

$$where\ F_1 \ldots F_K\ are\ K factors\ and\ R_1 \ldots R_N\ returns\ of\ the\ stocks\ 1..N$$

Let’s look into Python Example...

Summary

- Factor is a quantitative characteristic of stocks that tends to predict returns
- Each Factor needs to be normalised and adjusted to have good statistical properties
- A typical modern Quant Factor Strategy has 10+ factors
- Factors tend to be highly correlated and the art is how to combine the factors such that
  - avoid multicollinearity
  - capture as much information as possible
  - have a stable model with good out-of-sample performance
- Typical techniques to combine factors: stepwise regression, Gramm-Schmidt, Lasso/Ridge
- In the next two lectures we will cover a Neural Network and develop a python code to combine Quant factors using this Neural Network